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(54) **POWER SEMICONDUCTOR DEVICES**

(71) Applicant: **Samsung Electronics Co., Ltd.**, Suwon-si (KR)

(72) Inventors: **Taehun Kim**, Suwon-si (KR); **Doohyung Cho**, Suwon-si (KR); **Mingu Ko**, Suwon-si (KR); **Youngcheol Kim**, Suwon-si (KR); **Jongseob Kim**, Suwon-si (KR); **Gyeongseon Park**, Suwon-si (KR)

(73) Assignee: **Samsung Electronics Co., Ltd.**, Suwon-si (KR)

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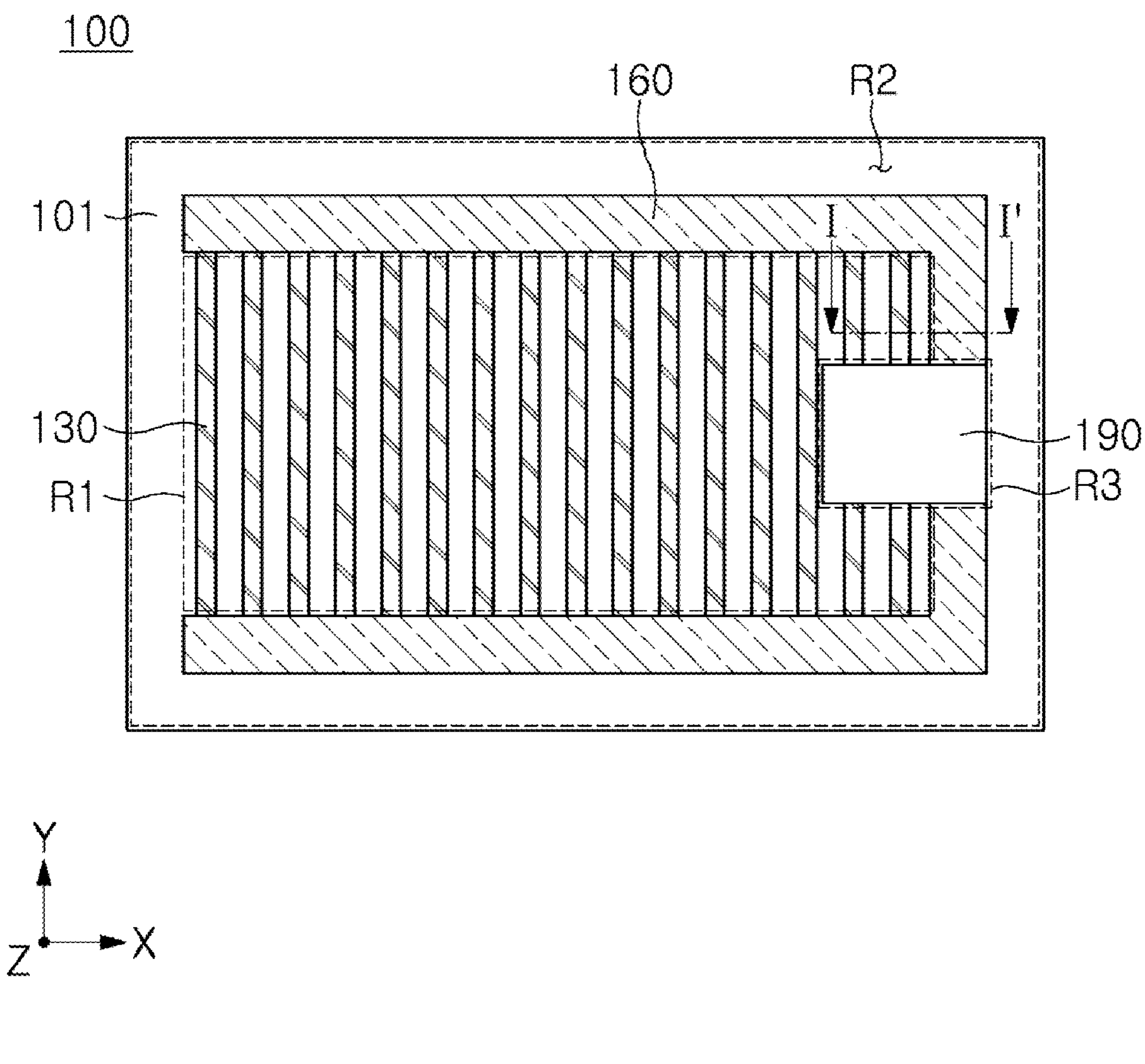
(57) **ABSTRACT**

A power semiconductor device includes a substrate including a cell region and an edge region extending from the cell region, a drift layer of a first conductivity type on the substrate, a first well region of a second conductivity type disposed within the drift layer, a source region of the first conductivity type disposed within the first well region, an insulating liner on the drift layer, gate electrodes spaced apart from each other on the insulating liner, an insulating pattern in the edge region, a conductive pattern spaced apart from the gate electrodes and covering a portion of the insulating liner and further covering a portion of the insulating pattern, a dielectric layer covering the gate electrodes and the conductive pattern, a source electrode on the dielectric layer, and a source contact plug penetrating through the dielectric layer and connecting the source electrode and the source region.



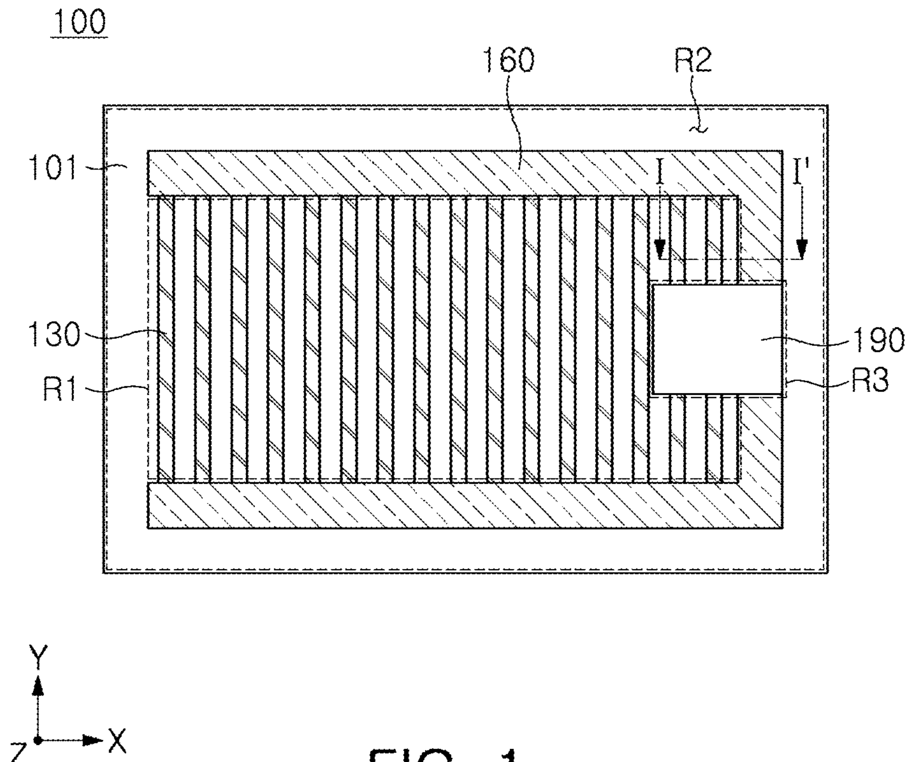
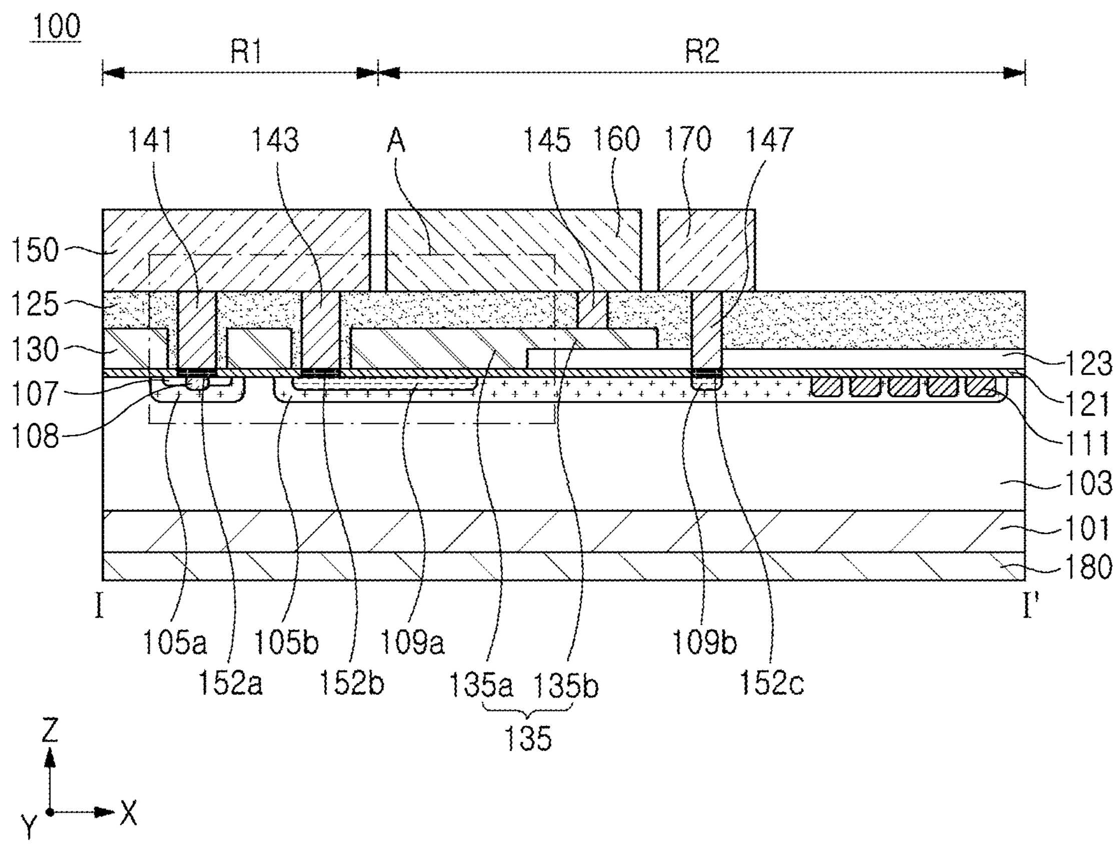


FIG. 1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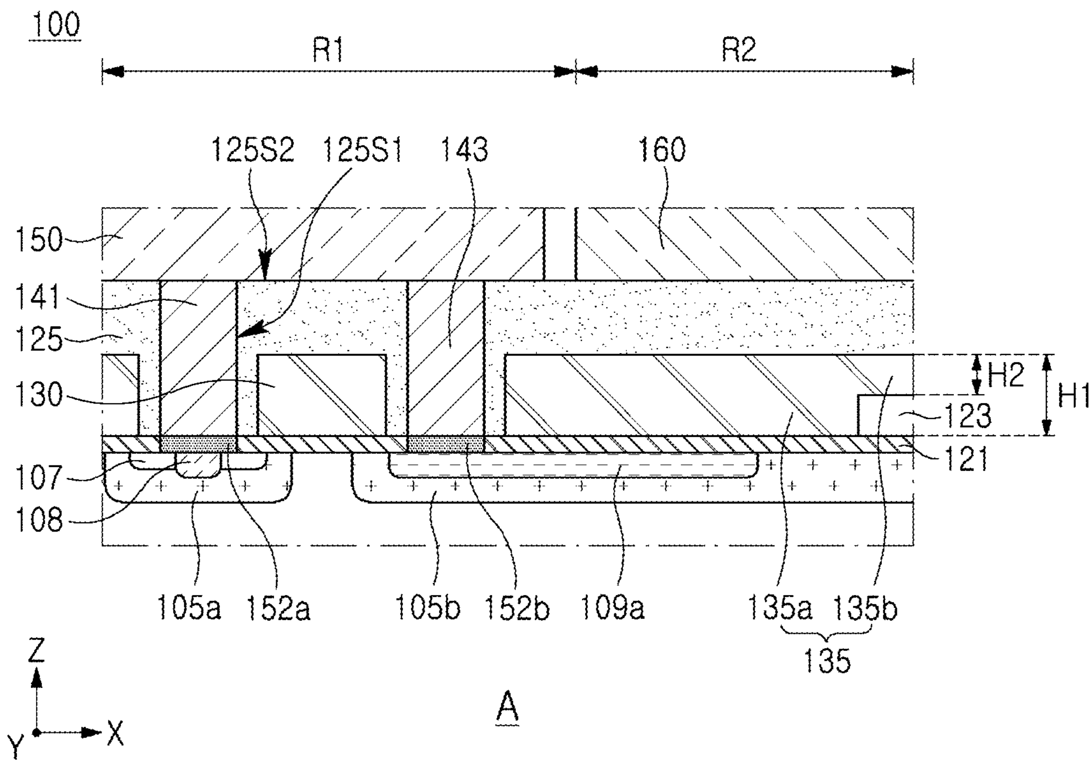
R2
141
143
A
145
160
170
147
150
125
130
107
108
123
121
111
103
101
180
I
I'
105a
152a
105b
152b
109a
135a
135b
135
109b
152c
Z
Y
X

FIG. 2A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R2
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125S1
143
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130
H2
H1
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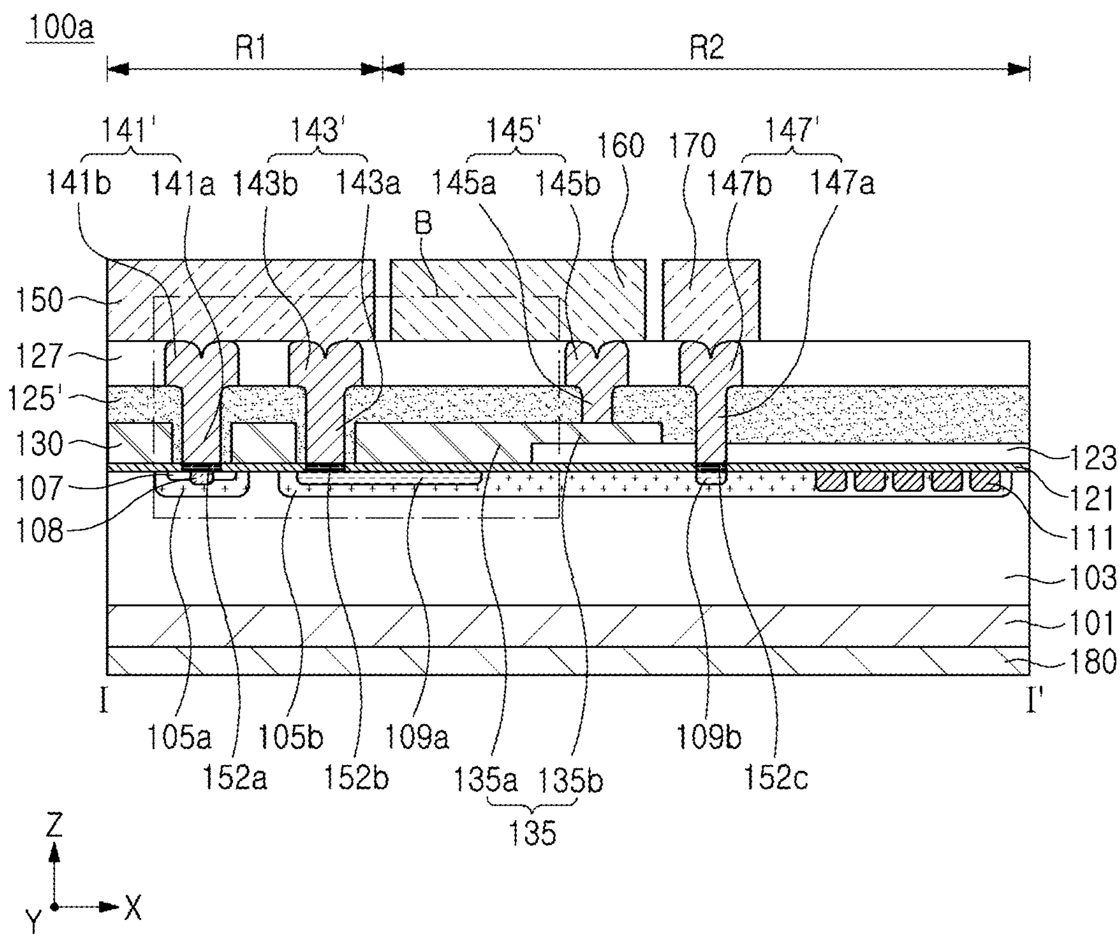
107
108
105a
152a
105b
152b
109a
135a
135b
135
Z
Y
X
A

FIG. 2B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R1
R2
141'
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147'
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143b
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145b
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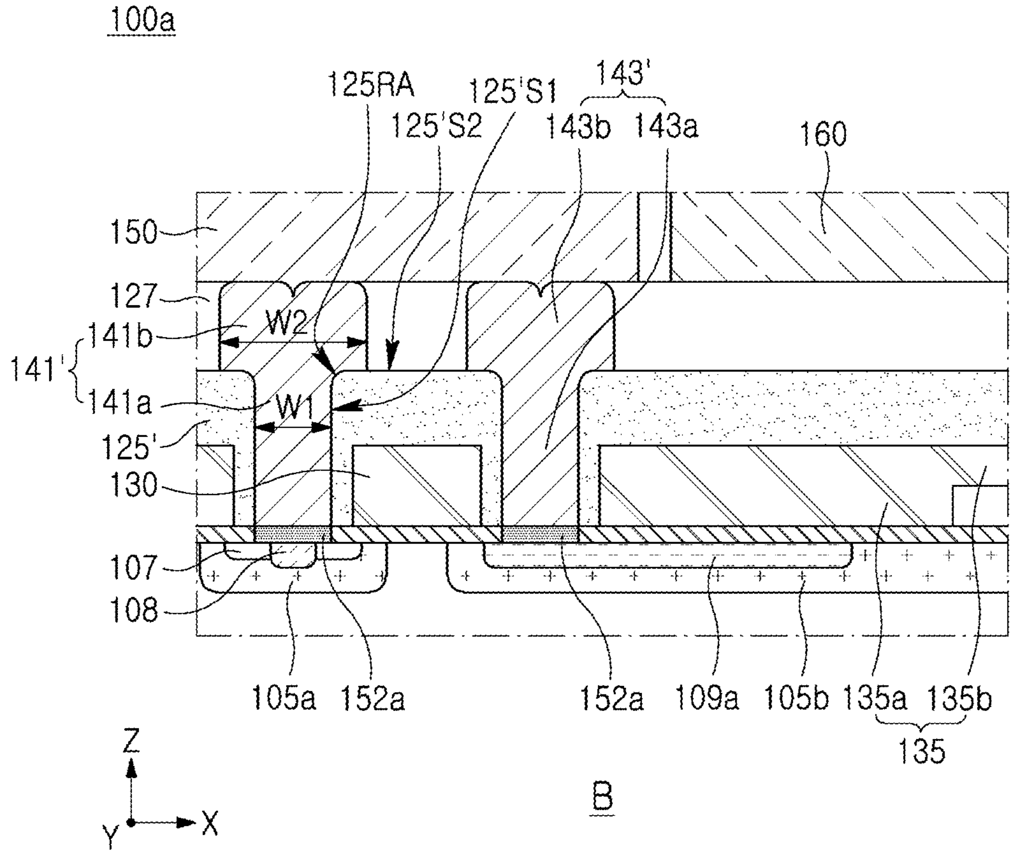
B
150
127
125'
130
107
108
123
121
111
103
101
180
I
I'
105a
152a
105b
152b
109a
135a
135b
135
109b
152c
Z
Y
X

FIG. 3A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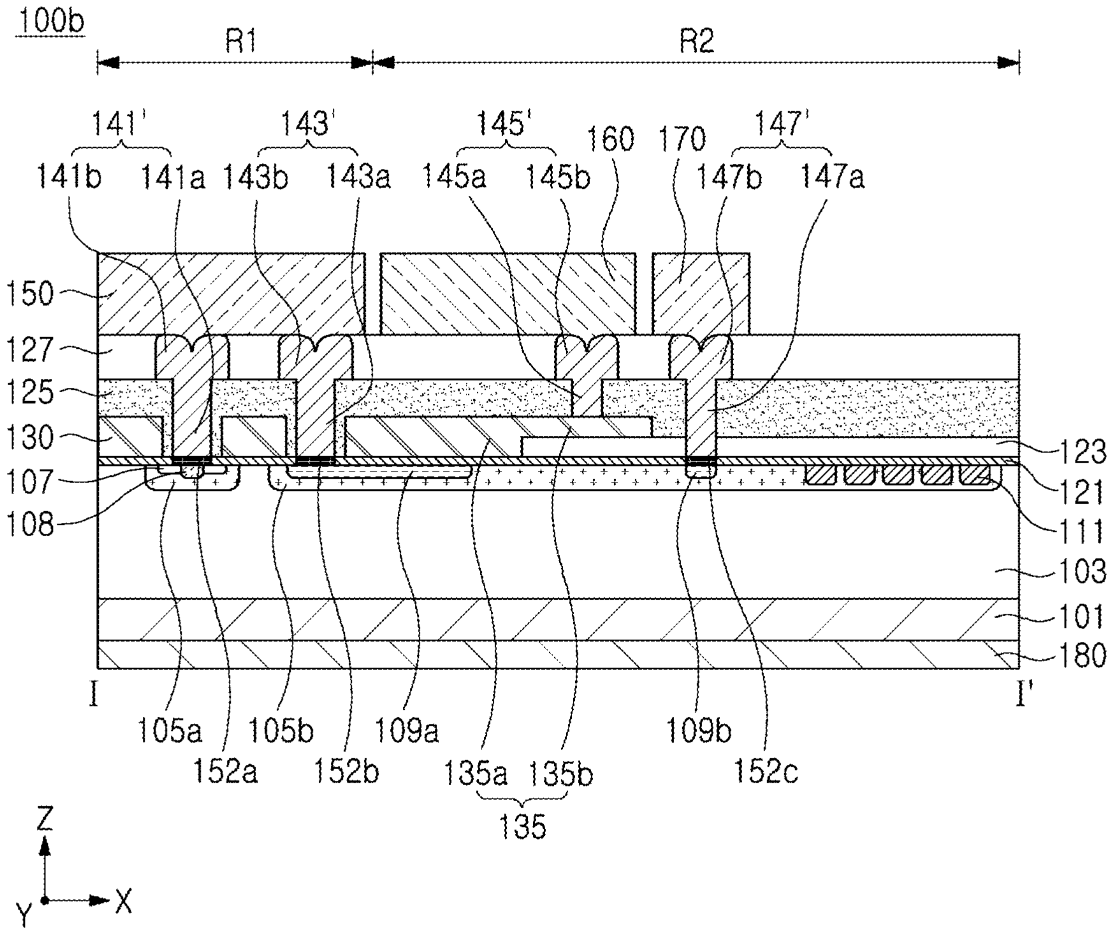
125'S1
143'
125'S2
143b
143a
160
150
127
141b
W2
141'
141a
W1
125'
130
107
108
105a
152a
152a
109a
105b
135a
135b
135
Z
Y
X
B

FIG. 3B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143a
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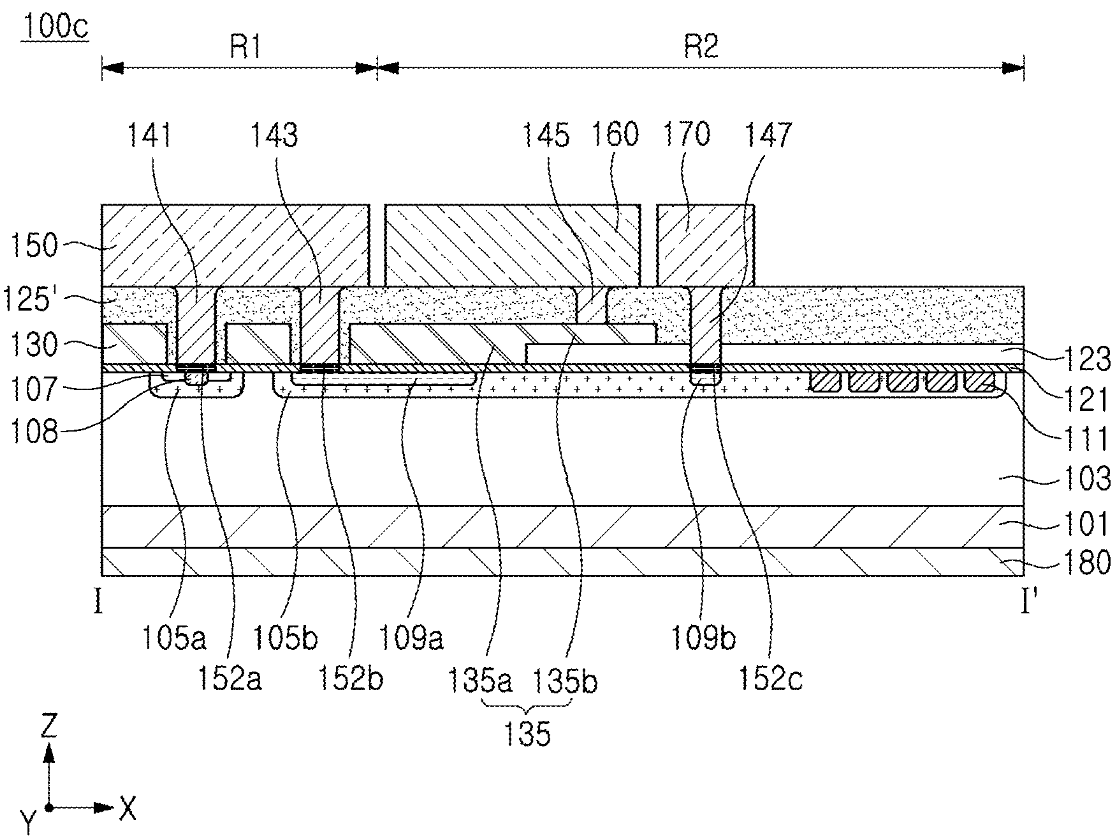
150
127
125
130
107
108
123
121
111
103
101
180
I
I'
105a
152a
105b
152b
109a
135a
135b
135
109b
152c
Z
Y
X

FIG. 4A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121
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103
101
180
I
I'
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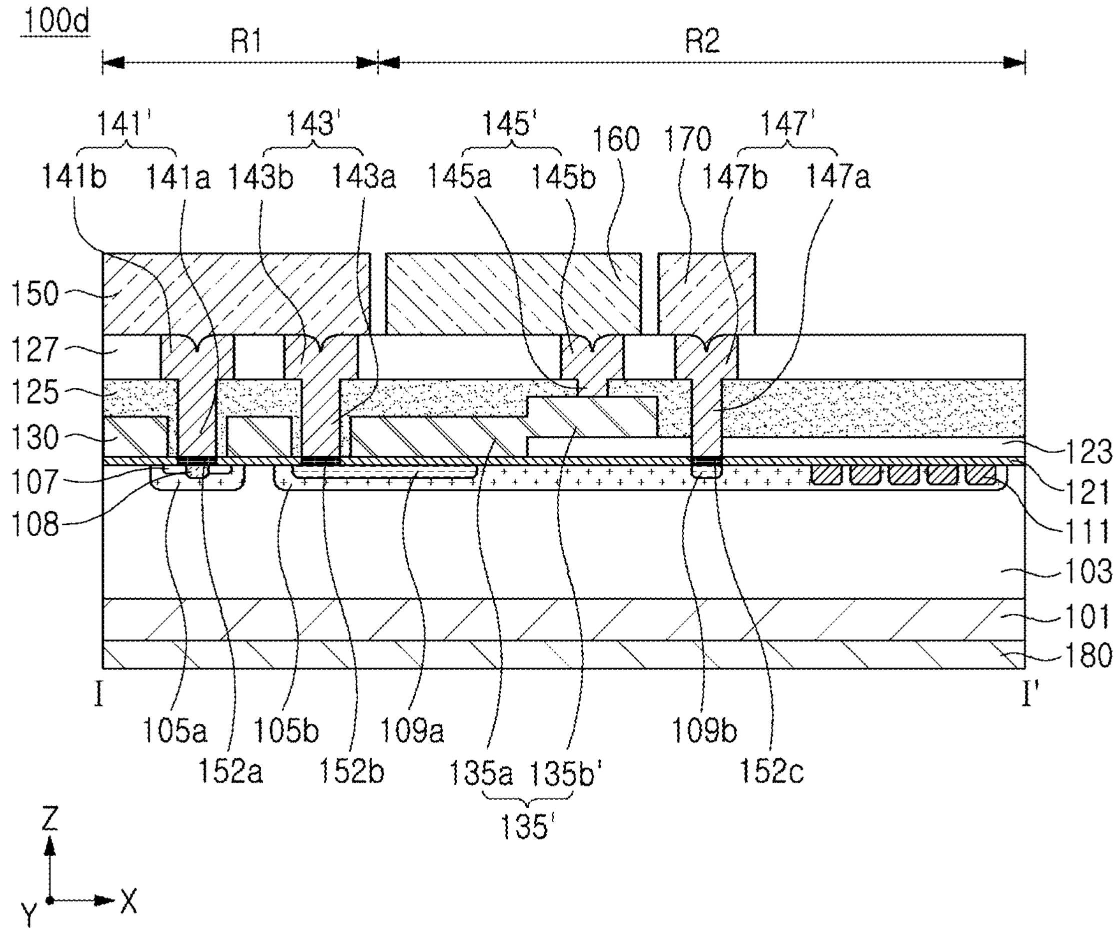
105b
152b
109a
135a
135b
135
109b
152c
Z
Y
X

FIG. 4B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147'
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143b
143a
145a
145b
160
170
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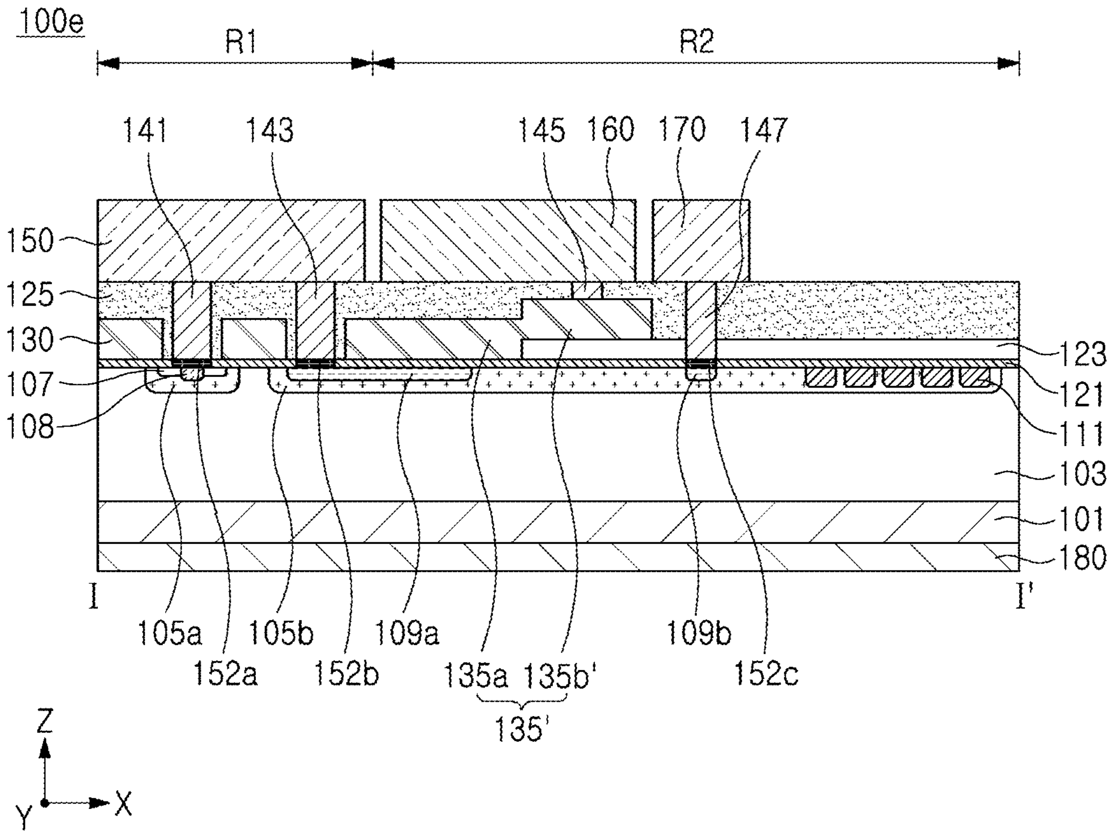
150
127
125
130
107
108
123
121
111
103
101
180
I
I'
105a
152a
105b
152b
109a
135a
135b'
135'
109b
152c
Z
Y
X

FIG. 4C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R2
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150
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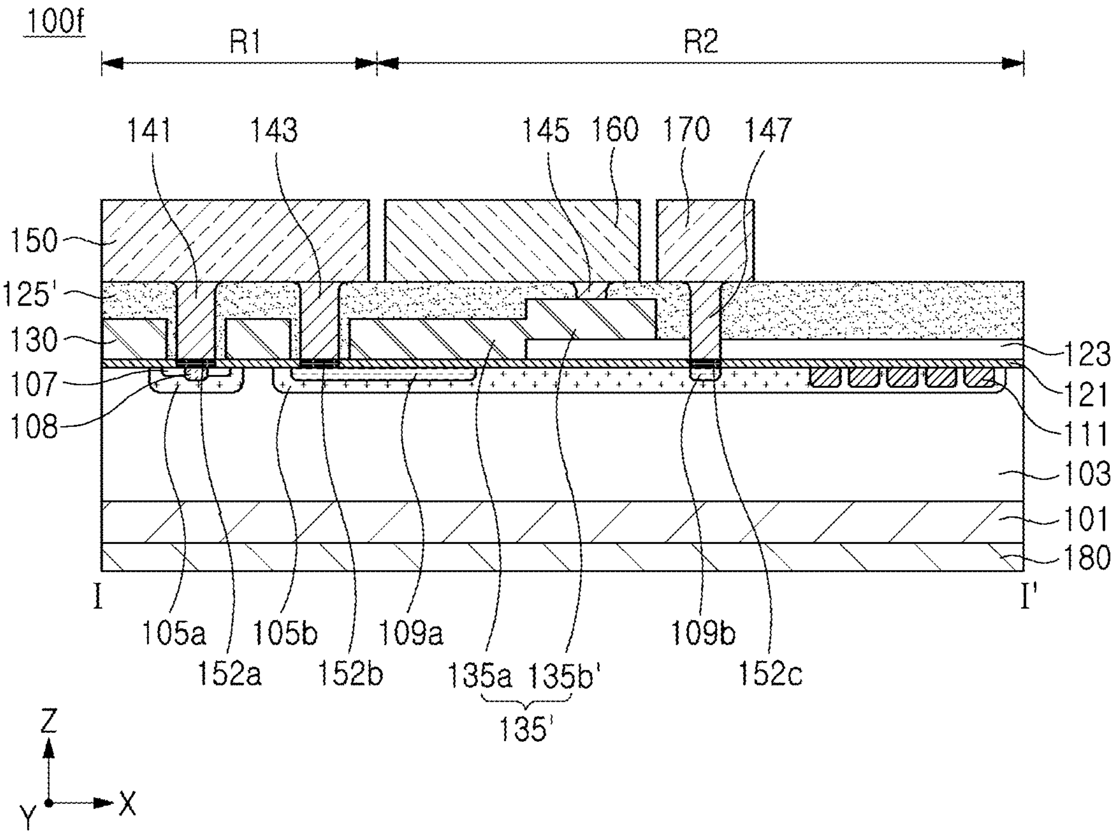
130
107
108
123
121
111
103
101
180
I
I'
105a
152a
105b
152b
109a
135a
135b'
135'
109b
152c
Z
Y
X

FIG. 4D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R2
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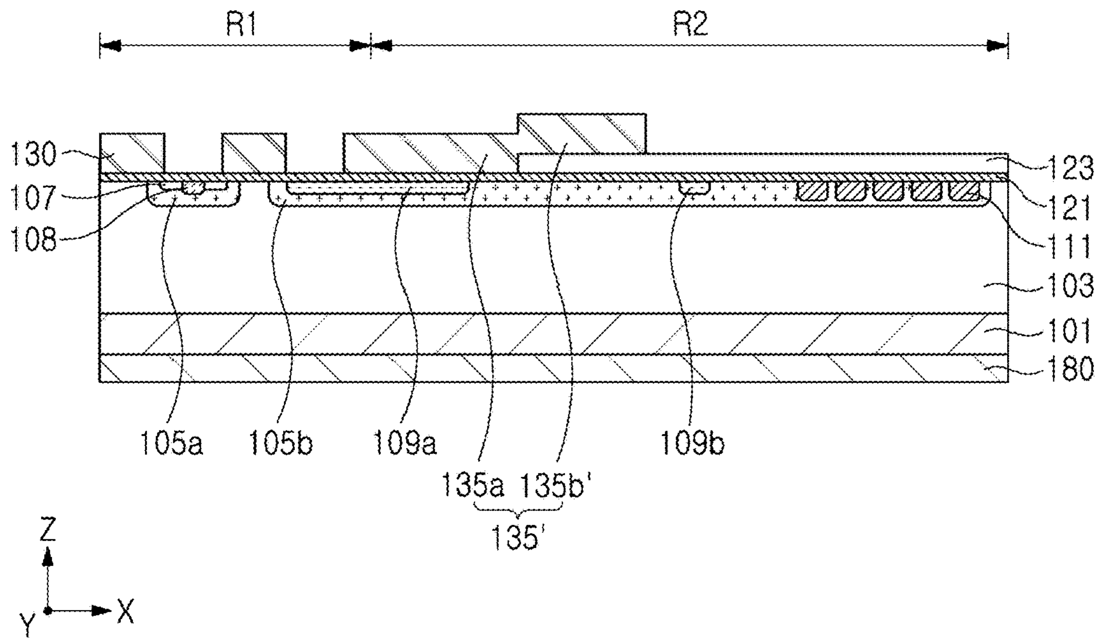
160
170
147
150
125'
130
107
108
123
121
111
103
101
180
I
I'
105a
152a
105b
152b
109a
135a
135b'
135'
109b
152c
Z
Y
X

FIG. 4E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R2
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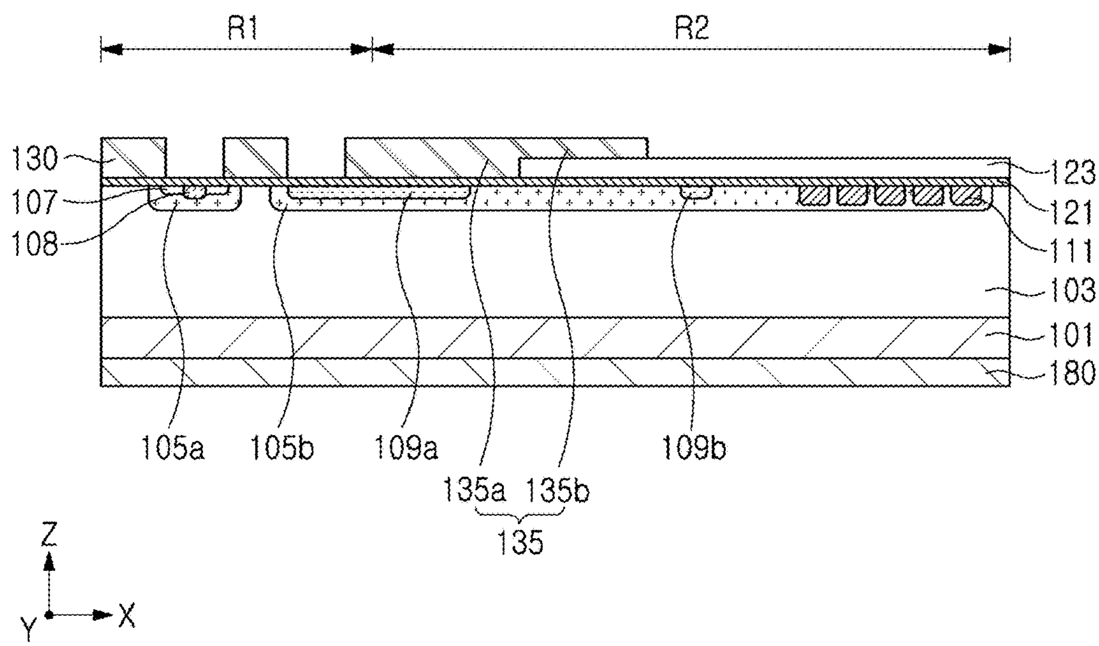
107
108
123
121
111
103
101
180
105a
105b
109a
109b
135a
135b'
135'
Z
Y
X

FIG. 5A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R2
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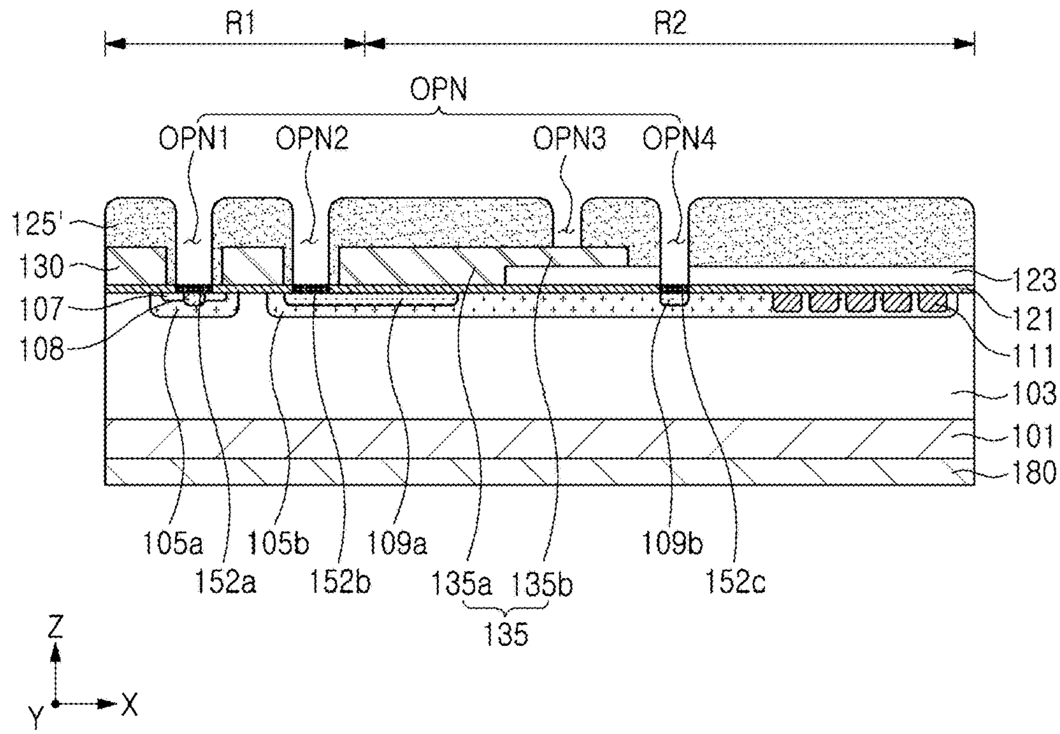
107
108
123
121
111
103
101
180
105a
105b
109a
109b
135a
135b
135
Z
Y
X

FIG. 5B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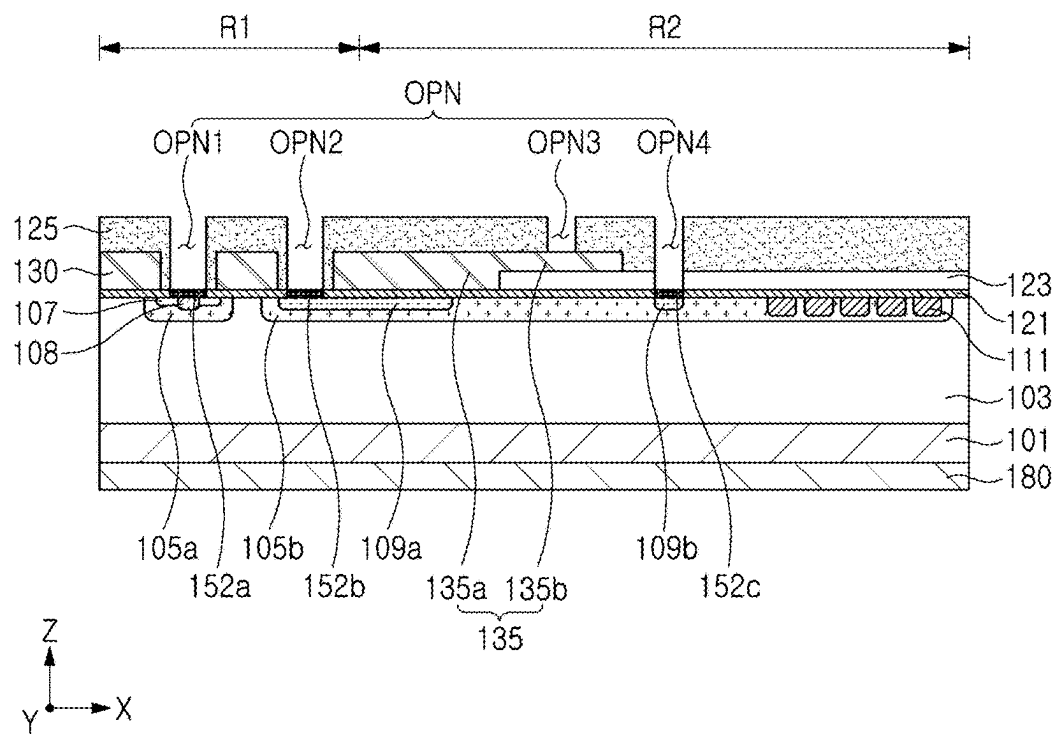
R2
OPN
OPN1
OPN2
OPN3
OPN4
125'
130
107
108
123
121
111
103
101
180
105a
152a
105b
152b
109a
135a
135b
135
109b
152c
Z
Y
X

FIG. 5C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OPN2
OPN3
OPN4
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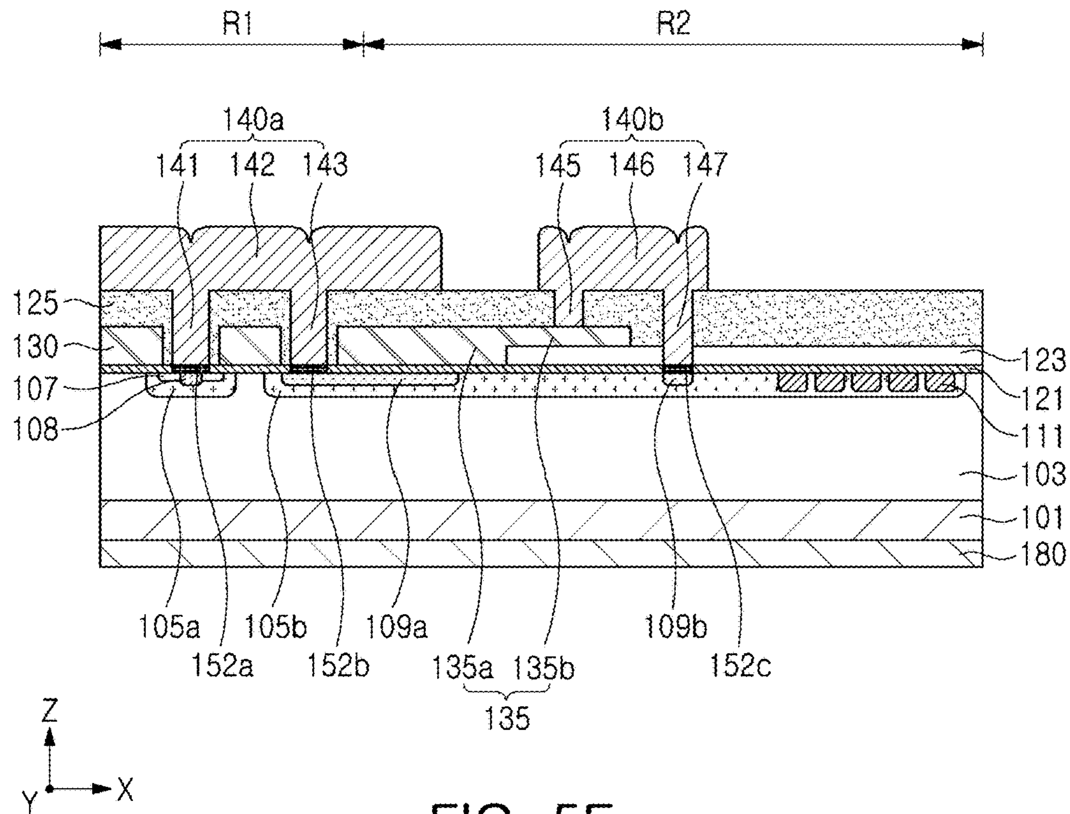
105b
152b
109a
135a
135b
135
109b
152c
Z
Y
X

FIG. 5D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R2
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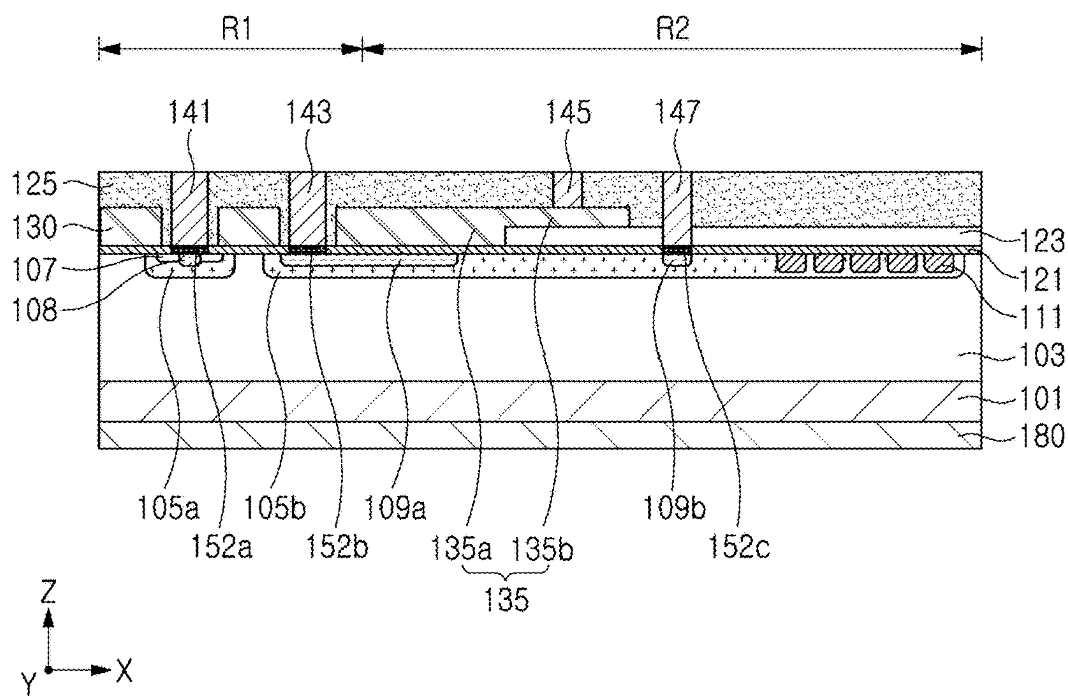
152c
Z
Y
X

FIG. 5E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105b
152b
109a
135a
135b
135
109b
152c
Z
Y
X

FIG. 5F

POWER SEMICONDUCTOR DEVICES

RELATED APPLICATION(S)

[0001] This application claims the benefit under 35 USC 119(a) of Korean Patent Application No. 10-2024-0033127 filed on Mar. 8, 2024 in the Korean Intellectual Property Office, the entire disclosure of which is incorporated herein by reference for all purposes.

FIELD

[0002] The present disclosure relate to power semiconductor devices, and to Metal Oxide Semiconductor Field Effect Transistor (MOSFET) power semiconductor devices.

BACKGROUND

[0003] Power semiconductor devices are semiconductor devices that operate in high voltage and high current environments and are used in fields requiring high power switching, such as power conversion, power converters, inverters, and the like. Power semiconductor devices are typically required to withstand high voltages, and now they are additionally required to perform high-speed switching operations. Accordingly, research into power semiconductor devices using Silicon Carbide (SiC), which has superior high voltage tolerance compared to silicon (Si), is being undertaken.

SUMMARY

[0004] Example embodiments provide a power semiconductor device having improved reliability by significantly reducing a step difference between a cell region and an edge region extending from the cell region.

[0005] According to example embodiments, a power semiconductor device includes a substrate including a cell region and an edge region extending from the cell region; a drift layer of a first conductivity type on the substrate; a first well region of a second conductivity type extending from an upper surface of the drift layer in the cell region and disposed within the drift layer; a source region of the first conductivity type extending from an upper surface of the first well region and disposed within the first well region; an insulating liner on the drift layer; gate electrodes spaced apart from each other on the insulating liner in the cell region, with the source region interposed therebetween; an insulating pattern in the edge region; a conductive pattern spaced apart from the gate electrodes and covering a portion of the insulating liner and a portion of the insulating pattern; a dielectric layer covering the gate electrodes and the conductive pattern; a source electrode on the dielectric layer; and a source contact plug penetrating through the dielectric layer and connecting the source electrode and the source region. The conductive pattern includes a first pattern in contact with the insulating liner and a second pattern in contact with the insulating pattern, and an upper surface of the first pattern is disposed on the same level as an upper surface of the second pattern.

[0006] According to example embodiments, a power semiconductor device includes a substrate including a first region and a second region surrounding the first region; a drift layer of a first conductivity type on the substrate; a first well region of a second conductivity type extending from an upper surface of the drift layer on the first region and disposed within the drift layer; a source region of the first conductivity type extending from an upper surface of the first well region and disposed within the first well region; an insulating liner on the drift layer; an insulating pattern on the second region; gate electrodes on the insulating liner on the first region and spaced apart from each other in a first direction with the source region interposed therebetween; a conductive pattern spaced apart from the gate electrodes and covering a portion of the insulating liner and a portion of the insulating pattern; a dielectric layer covering the gate electrodes and the conductive pattern; and a source electrode on the dielectric layer in the first region. The conductive pattern includes a first pattern disposed on the insulating liner and a second pattern extending from the first pattern and disposed on the insulating pattern. The first pattern has a first height in a vertical direction intersecting the first direction from an upper surface of the insulating liner, and the second pattern has a second height, lower than the first height, in the vertical direction, from an upper surface of the insulating pattern.

[0007] According to example embodiments, a power semiconductor device includes a substrate of a first conductivity type; a drift layer of a first conductivity type on the substrate; a first well region of a second conductivity type on the drift layer; a second well region of the second conductivity type disposed on the drift layer and spaced apart from the first well region in a first direction; source regions of the first conductivity type on the first well region; an insulating liner on the drift layer; gate electrodes spaced apart from each other, with the source region interposed therebetween, on the first well region; an insulating pattern disposed on a portion of the insulating liner, on the second well region; a conductive pattern spaced apart from the gate electrodes and covering a portion of the insulating liner and a portion of the insulating pattern on the second well region; a dielectric layer covering the gate electrodes and the conductive pattern; a source electrode on the dielectric layer; and a source contact plug penetrating through the dielectric layer and connecting the source electrode and the source region. The conductive pattern includes a first pattern in contact with the insulating liner and a second pattern in contact with the insulating pattern, and an upper surface of the first pattern and an upper surface of the second pattern are disposed on the same level. An upper surface of the dielectric layer is disposed on the same level as an upper surface of the source contact plug.

BRIEF DESCRIPTION OF DRAWINGS

[0008] The above and other aspects, features, and advantages of the present disclosure will be more clearly understood from the following detailed description, taken in conjunction with the accompanying drawings, in which:

[0009] FIG. **1** is a schematic plan view of a power semiconductor device according to example embodiments;

[0010] FIG. **2**A is a cross-sectional view illustrating an example embodiment of the power semiconductor device of FIG. **1**, taken along line I-I';

[0011] FIG. **2**B is an enlarged view of region A of the power semiconductor device of FIG. **2**A;

[0012] FIG. **3**A is a cross-sectional view illustrating another embodiment of the power semiconductor device of FIG. **1**, taken along line I-I';

[0013] FIG. **3**B is an enlarged view of region B of the power semiconductor device of FIG. **3**A;

[0014] FIGS. **4**A to **4**E are cross-sectional views illustrating another embodiment of the power semiconductor device of FIG. **1**, taken along line I-I'; and

[0015] FIGS. **5**A to **5**F are diagrams illustrating an example embodiment of a method of manufacturing the power semiconductor device of FIG. **2**A.

DETAILED DESCRIPTION

[0016] Hereinafter, example embodiments of the present disclosure will be described in more detail with reference to the attached drawings. The same reference numerals are used for the same components in the drawings, and duplicate descriptions of the same components are omitted.

[0017] FIG. **1** is a schematic plan view of a power semiconductor device according to example embodiments. FIG. **2**A is a cross-sectional view illustrating an example embodiment of the power semiconductor device of FIG. **1**, taken along line I-I'. FIG. **2**B is an enlarged view of region A of the power semiconductor device of FIG. **2**A.

[0018] Referring to FIGS. **1** and **2**A, a power semiconductor device **100** may include a substrate **101**, a drift layer **103** on the substrate **101**, first and second well regions **105***a* and **105***b* extending from the upper surface of the drift layer **103** and spaced apart from each other, source regions **107** disposed on the first well region **105***a*, a well contact region **108** disposed on one side of the source regions **107**, peripheral contact regions **109***a* and **109***b* disposed on the second well region **105***b*, an insulating liner **121** disposed on the drift layer **103**, gate electrodes **130** on insulating liner **121**, a conductive pattern **135** spaced apart from the gate electrodes **130**, a dielectric layer **125** covering the gate electrodes **130** and the conductive pattern **135**, a source electrode **150** and a gate bus line **160** on the dielectric layer **125**, a source contact plug **141** penetrating through the dielectric layer **125** and connecting to the source regions **107** and the source electrode **150**, and a drain electrode **180** on the lower surface of the substrate **101**.

[0019] The substrate **101** may have an upper surface extending in a first direction (X-direction) and a second direction (Y-direction). The substrate **101** may include a semiconductor material, for example, SiC. However, in some embodiments, the substrate **101** may include a group IV semiconductor material such as Si or Ge, or a compound semiconductor material such as SiGe, GaAs, InAs, or InP.

[0020] The substrate **101** may be provided as a bulk wafer or an epitaxial layer. The substrate **101** may include first conductivity type impurities and thus may have a first conductivity type. In some embodiments, the first conductivity type may be, for example, N-type, and the first conductivity type impurities may be N-type impurities, for example, nitrogen (N) and/or phosphorus (P). In some embodiments, the first conductivity type may be, for example, P-type, and the first conductivity type impurities may be, for example, P-type impurities such as aluminum (Al).

[0021] The substrate **101** may include a first region R**1** where gate electrodes **130** are disposed, a second region R**2** surrounding at least one side of the first region R**1** and in which gate bus lines **160** are disposed, and a third region R**3** where the gate pad **190** electrically connected to the gate electrodes **130** is disposed. In this document, the first region R**1** may be referred to as a cell region, and the second region R**2** may be referred to as an edge region. The third region R**3** may be referred to as a gate pad region.

[0022] The drift layer **103** may be disposed on the substrate **101**. The drift layer **103** may include a semiconductor material. For example, the semiconductor material may include SiC. The drift layer **103** may be an epitaxial layer grown on the substrate **101**. The drift layer **103** may include impurities of a first conductivity type and thus may have the first conductivity type. The concentration of first conductivity type impurities in the drift layer **103** may be lower than the concentration of first conductivity type impurities in the substrate **101**. In example embodiments, the first conductivity type impurities in the substrate **101** and the drift layer **104** may be the same or different from each other.

[0023] The first and second well regions **105***a* and **105***b* may be disposed at a predetermined depth from the upper surface of the drift layer **103**. The first and second well regions **105***a* and **105***b* may be disposed to be spaced apart from each other in the horizontal direction. The first and second well regions **105***a* and **105***b* may include a semiconductor material, for example, SiC. The first and second well regions **105***a* and **105***b* may be regions having a second conductivity type and may include impurities of the second conductivity type. For example, the second conductivity type may be P-type, and the second conductivity type impurities may be P-type impurities such as aluminum (Al) or boron (B).

[0024] The first well region **105***a* may overlap the first region R**1** of the substrate **101** in the vertical direction (Z-direction). The second well region **105***b* may be spaced apart from the first well region **105***a* in the first direction (X-direction) and may be disposed to surround at least one side of the first well region **105***a*. In an example, the second well region **105***b* may extend from the first region R**1** to the second region R**2**. The second well region **105***b* is illustrated as being disposed across the first and second regions R**1** and R**2**, but is not limited thereto. For example, the second well region **105***b* may not overlap the first region R**1** in the vertical direction (Z-direction), but may overlap the second region R**2** in the vertical direction (Z-direction).

[0025] The first and second well regions **105***a* and **105***b* may have substantially the same thickness in the vertical direction (Z-direction). However, the present disclosure is not limited thereto, and the thickness of the first well region **105***a* in the vertical direction (Z-direction) may be smaller or larger than the thickness of the second well region **105***b*.

[0026] The source regions **107** may each be disposed in the first well region **105***a* and may be disposed at a predetermined depth from the upper surface of the first well region **105***a*. The thickness of the source region **107** in the vertical direction (Z-direction) may be smaller than the thickness of the first well region **105***a*. Source region **107** may include a semiconductor material, for example, SiC. The source region **107** may be a region having a first conductivity type and may include the first conductivity type impurities described above. The concentration of impurities of the first conductivity type in the source region **107** may be higher than the concentration of impurities of the first conductivity type in the drift layer **103**, but is not limited thereto.

[0027] The power semiconductor device **100** according to example embodiments may further include a well contact region **108** disposed on one side of the source regions **107**. The well contact region **108** may be disposed at a predetermined depth from the upper surface of the drift layer **103**. In an example, the well contact region **108** may extend from the upper surface of the drift layer **103** and be disposed within the first well region **105***a*. The well contact region **108** may be disposed and/or formed across the source region **107** in the vertical direction (Z-direction). In an example, the well contact region **108** may be disposed between the source regions **107** within the first well region **105***a* to apply the voltage from the source electrode **150** to the first well region **105***a*. The well contact region **108** may include a semiconductor material, for example, SiC. The well contact region **108** may be a region having the second conductivity type, and may include the second conductivity type impurities described above. The concentration of the second conductivity type impurities in the well contact region **108** may be higher than the concentration of the second conductivity type impurities in the first well region **105***a*.

[0028] The peripheral contact regions **109***a* and **109***b* may extend from the upper surface of the drift layer **103** and be disposed within the second well region **105***b*. The peripheral contact regions **109***a* and **109***b* may be spaced apart from the well contact region **108** in a horizontal direction (for example, first and second directions (X- and Y-directions)). In an example, the peripheral contact regions **109***a* and **109***b* may include a first peripheral contact region **109***a* and a second peripheral contact region **109***b* spaced apart from the first peripheral contact region **109***a*. The first peripheral contact region **109***a* may be disposed to extend from within the second well region **105***b* disposed in the first region R**1** to the second region R**2**. The concentration of second conductivity type impurities in the peripheral contact regions **109***a* and **109***b* may be higher than the concentration of second conductivity type impurities in the second well region **105***b*. In an example, the peripheral contact regions **109***a*, **109***b* may be surface depletion protection regions to prevent depletion in the region at or near the upper surface of the drift layer **103** disposed on the second region R**2**.

[0029] The first and second peripheral contact regions **109***a* and **109***b* may include a semiconductor material, for example, SiC. The first and second peripheral contact regions **109***a* and **109***b* may be regions having the second conductivity type and may include the above-described second conductivity type impurities. The concentration of second conductivity type impurities in the first and second peripheral contact regions **109***a* and **109***b* may be higher than the concentration of second conductivity type impurities in the second well region **105***b*.

[0030] The power semiconductor device **100** according to example embodiments may further include at least one guard ring **111** disposed in the second well region **105***b* disposed in the second region R**2**. The at least one guard ring **111** is disposed to be spaced apart from the conductive pattern **135**, and may be disposed to surround the first region R**1** in the area closest to the edge of the substrate **101** within the second region R**2**. At least one guard ring **111** may be a region having a second conductivity type and may include second conductivity type impurities. The concentration of impurities of the second conductivity type in the at least one guard ring **111** may be higher than the concentration of impurities of the second conductivity type in the second well region **105***b*. The drawing illustrates five guard rings **111**, but is not limited thereto.

[0031] The gate electrodes **130** may be disposed on the first region R**1** of the substrate **101**. The gate electrodes **130** may extend in the second direction (Y-direction) on the first region R**1** and be spaced apart from each other in the first direction (X-direction). The gate electrodes **130** may be disposed on the insulating liner **121** disposed on the first region R**1** of the substrate **101**. The gate electrode **130** may overlap one end of the source regions **107** and the first well region **105***a* outside the source regions **107** in the vertical direction (Z-direction). The gate electrodes **130** may be spaced apart from the source region **107**, the first well region **105***a*, and the drift layer **103** by the insulating liner **121**.

[0032] When viewed in plan view, the gate electrodes **130** are illustrated as not overlapping with the gate bus lines **160** and the gate pad **190**, but some of the gate electrodes **130** may be disposed to overlap the gate bus lines **160** and the gate pad **190** in the vertical direction (Z-direction).

[0033] The gate electrode **130** may include a conductive material, for example, semiconductor materials such as doped polycrystalline silicon, metal nitride, such as titanium nitride (TiN), tantalum nitride (TaN), or tungsten nitride (WN), and/or metallic substances such as aluminum (Al), tungsten (W), or molybdenum (Mo). Depending on example embodiments, the gate electrode **130** may be composed of two or more layers.

[0034] The insulating liner **121** may be disposed on the upper surface of the drift layer **103**. The insulating liner **121** may extend onto the source region **107**, the first well region **105***a* outside of the source region **107**, the drift layer **103**, and the second well region **105***b*. In an example, the insulating liner **121** may function as a gate insulating layer in the first region R**1**.

[0035] The insulating liner **121** may include oxide or nitride. For example, the insulating liner **121** may include silicon oxide. The insulating liner **121** may include a high-k material. A high dielectric constant material may refer to a dielectric material having a higher dielectric constant than a silicon oxide film (SiO_2). For example, the high dielectric constant material may be any one of aluminum oxide (Al_2O_3), tantalum oxide (Ta_2O_3), titanium oxide (TiO_2), yttrium oxide (Y_2O_3), zirconium oxide (ZrO_2), zirconium silicon oxide ($ZrSi_xO_y$), hafnium oxide (HfO_2), hafnium silicon oxide ($HfSi_xO_y$), Lanthanum oxide (La_2O_3), lanthanum aluminum oxide ($LaAl_xO_y$), lanthanum hafnium oxide ($LaHf_xO_y$), hafnium aluminum oxide ($HfAl_xO_y$), and praseodymium oxide (Pr_2O_3). The insulating liner **121** may be composed of a plurality of insulating films, and each of the plurality of insulating films may include one of the above-mentioned materials.

[0036] The gate pad **190** may be disposed/formed on the third region R**3** on the substrate **101**. In an example, the third region R**3** may be adjacent to the first region R**1** in the first direction (X-direction). In an example, the gate pad **190** may be disposed on one side of the gate electrodes **130** and electrically connected to the gate electrodes **130** through gate bus lines **160**. The gate pad **190** is electrically connected to the pad metal layer disposed thereon and may receive an electrical signal through the pad metal layer. In an example, some of the gate electrodes **130** may extend to a lower portion of the gate pad **190** and may overlap and be electrically connected to the gate pad **190** in the vertical direction (Z-direction). Depending on example embodiments, the shape of the gate pad **190** may be a square, circular, or oval shape in a plan view.

[0037] The gate bus lines **160** may be disposed and/or formed on the second region R**2** on the substrate **101**. The second region R**2** may include regions surrounding one side extending in the first direction (X-direction) of the first region R**1** and the other side parallel to one side of the first region R1. In an example, the second region R2 may be an area surrounding the first region R1 and the periphery of the third region R3 disposed on one side of the first region R1.

[0038] The power semiconductor device **100** may include a plurality of gate bus lines **160**. For example, the power semiconductor device may include two gate bus lines **160**. The gate bus lines **160** may connect one end of each of the gate electrodes **130** and the other end opposite to one end of each of the gate electrodes **130**. In an example, the gate bus lines **160** may be disposed symmetrically about the X-axis with the gate pad **190** as the center. In an example, the gate bus lines **160** may include a first part connected to the gate pad **190** and extending in the second direction (Y-direction), and a second part extending from the first part in a first direction (X-direction) intersecting the second direction (Y-direction), which is the extension direction of the gate electrodes **130**. The gate bus lines **160** may be connected to the conductive pattern **135** in the vertical direction (Z-direction) through the gate contact plug **145** penetrating through the dielectric layer **125**.

[0039] The gate pad **190** and the gate bus lines **160** may include a conductive material, for example, a metal material. The gate pad **190** and the gate bus lines **160** may include at least one of, for example, titanium nitride (TiN), titanium (Ti), titanium carbide (TIC), tantalum nitride (TaN), tungsten nitride (WN), aluminum (Al), tungsten (W), and molybdenum (Mo). In an example, the gate pad **190** and the gate bus lines **160** may include the same material or different materials.

[0040] The insulating pattern **123** may be disposed on a portion of the insulating liner **121** disposed on the second region R2. The insulating pattern **123** may overlap the second well region **105***b* disposed on the second region R2 in the vertical direction (Z-direction). The insulating pattern **123** may be disposed in a ring shape to surround the first region R1 on the second region R2. However, the present disclosure is not limited thereto. For example, the insulating pattern **123** may be disposed in the form of an open curved line. In an example, the upper surface of the insulating pattern **123** may be disposed at a lower level than the upper surface of the gate electrode **130**. However, the present disclosure is not limited thereto, and the upper surface of the insulating pattern **123** may be disposed on substantially the same level as the upper surface of the gate electrode **130**.

[0041] The insulating pattern **123** may include an insulating material, and for example, may include oxide or nitride. The insulating pattern **123** may include a silicon oxide film (SiO_2).

[0042] The conductive pattern **135** may be disposed on the insulating liner **121** and the insulating pattern **123** to be spaced apart from the gate electrodes **130**. In an example, the conductive pattern **135** may be disposed on a region of the insulating liner **121** and the insulating pattern **123** between the gate electrodes **130** and the insulating pattern **123**. In an example, the conductive pattern **135** may include a first pattern **135***a* overlapping the insulating liner **121** in the vertical direction (Z-direction) and a second pattern **135***b* extending from the first pattern **135***a* and overlapping the insulating pattern **123**. In an example, the first pattern **135***a* may contact the upper surface of the insulating liner **121** disposed on the second well region **105***b*. The second pattern **135***b* covers a portion of the insulating pattern **123** and may be in contact with the upper surface of the portion of the insulating pattern **135***b*.

[0043] The upper surface of the conductive pattern **135** may be disposed on the same level as the upper surface of the gate electrodes **130**. In an example, the upper surface of the first pattern **135***a* and the upper surface of the second pattern **135***b* may be disposed on the same level.

[0044] The first pattern **135***a* may have a first height H1 in the vertical direction (Z-direction) on the insulating liner **121** disposed on the second well region **105***b*. The second pattern **135***b* may have a second height H2 that is shorter than the first height H1 in the vertical direction (Z-direction) on the insulating pattern **123**. In an example, the gate electrodes **130** may have a height equal to the first height H1 in the vertical direction (Z-direction) on the insulating liner **121** disposed on the first well region **105***a*.

[0045] A portion of the first pattern **135***a* may overlap the first peripheral contact region **109***a* in the second well region **105***b* in the vertical direction (Z-direction).

[0046] The conductive pattern **135** may include a conductive material, for example, a semiconductor material such as doped polycrystalline silicon, a metal nitride, such as titanium nitride (TiN), tantalum nitride (TaN), or tungsten nitride (WN), and/or a metallic substance such as aluminum (Al), tungsten (W), or molybdenum (Mo). Depending on example embodiments, the conductive pattern **135** may be composed of two or more layers. In an example, the conductive pattern **135** may include the same material as the gate electrodes **130**. However, the present disclosure is not limited thereto, and the conductive pattern **135** may include a conductive material different from that of the gate electrodes **130**.

[0047] The dielectric layer **125** may cover a portion of the insulating liner **121**, the gate electrodes **130**, and the conductive pattern **135**. The dielectric layer **125** may include an insulating material and may include at least one of silicon oxide, silicon nitride, and silicon oxynitride.

[0048] A source electrode **150** and a gate bus line **160** may be disposed on the dielectric layer **125**. The source electrode **150** may be disposed on the first region R1 of the substrate **101**, and the gate bus line **160** may be disposed on the second region R2 and spaced apart from the source electrode **150** in the first direction (X-direction).

[0049] The source electrode **150** may be disposed on the dielectric layer **125** and may be electrically connected to the source regions **107** and the well contact region **108** through the source contact plug **141**. The source electrode **150** may be disposed on the dielectric layer **125** and electrically connected to the first peripheral contact region **109***a* through the first peripheral contact plug **143**.

[0050] The source electrode **150** may include a metallic substance, for example, contain at least one of nickel (Ni), aluminum (Al), titanium (Ti), silver (Ag), vanadium (V), tungsten (W), cobalt (Co), molybdenum (Mo), copper (Cu), and ruthenium (Ru).

[0051] The source contact plug **141** may penetrate the dielectric layer **125** and be disposed between the source electrode **150** and the source regions **107** of the first well region **105***a*. In an example, the source contact plug **141** may be disposed between the gate electrodes **130**. The source contact plug **141** may include a conductive plug. The conductive plug may include a metal material, for example, tungsten (W), titanium (Ti), aluminum (Al), or copper (Cu).

[0052] A first metal-semiconductor compound layer **152***a* may be disposed at the interface in contact with the source contact plug **141**, the source regions **107**, and the well contact region **108**.

[0053] The dielectric layer **125** may include a side **125S1** surrounding the side surface of the source contact plug **141** and an upper surface **125S2** extending from the side **125S1** and contacting the source electrode **150**. The dielectric layer **125** may include a corner formed in an area where the side **125S1** and the upper surface **125S2** contact each other.

[0054] The dielectric layer **125** may expose the upper surface of the source contact plug **141**. In an example, the upper surface of the source contact plug **141** may be disposed on substantially the same level as the upper surface of the dielectric layer **125**.

[0055] The gate bus line **160** may be connected to the conductive pattern **135** through the gate contact plug **145**. The gate contact plug **145** may connect the gate bus line **160** and the conductive pattern **135** by penetrating through the dielectric layer **125**. The gate contact plug **145** may contact the upper surface of the second pattern **135***b* of the conductive pattern **135**.

[0056] The dielectric layer **125** may expose the upper surface of the gate contact plug **145**. In an example, the upper surface of the gate contact plug **145** may be disposed on the same level as the upper surface of the dielectric layer **125**. In an example, the upper surface of the gate contact plug **145** may be disposed on the same level as the upper surface of the source contact plug **141**.

[0057] The power semiconductor device **100** according to example embodiments may further include a conductive terminal **170** disposed on the dielectric layer **125**, a first peripheral contact plug **143** disposed between the gate electrode **130** and the conductive pattern **135**, and a second peripheral contact plug **147** disposed between the conductive terminal **170** and the second well region **105***b*.

[0058] The first peripheral contact plug **143** may penetrate the dielectric layer **125** to connect the source electrode **150** and the first peripheral contact region **109***a* of the second well region **105***b*. The second peripheral contact plug **147** may connect the conductive terminal **170** and the second peripheral contact region **109***b* by penetrating through the dielectric layer **125** and the insulating pattern **123**.

[0059] A second metal-semiconductor compound layer **152***b* may be disposed at an interface that contacts the first peripheral contact plug **143** and the first peripheral contact region **109***a*. A third metal-semiconductor compound layer **152***c* may be disposed at an interface that contacts the second peripheral contact plug **147** and the second peripheral contact region **109***b*.

[0060] The metal-semiconductor compound layers **152***a*, **152***b*, and **153***c* may include a metal element and a semiconductor element, for example, at least one of TiSi, CoSi, MoSi, LaSi, NiSi, TaSi, or Wsi.

[0061] The upper surfaces of the first and second peripheral contact plugs **143** and **147** may be disposed on the same level as the upper surface of the dielectric layer **125**.

[0062] The upper surfaces of the source contact plug **141**, the gate contact plug **145**, and the first and second peripheral contact plugs **143** and **147** may be disposed on the same level.

[0063] The source electrode **150**, gate bus line **160**, and conductive terminal **170** may be in contact with the upper surface of the dielectric layer **125**.

[0064] The lower surface of the source electrode **150** and the lower surface of the gate bus line **160** may be disposed on the same level.

[0065] The power semiconductor device **100** according to example embodiments may include a conductive pattern **135** having an upper surface disposed on the same level as the upper surface of the gate electrodes **130** disposed in the first region R**1** and extending from the first region R**1** and disposed in the second region R**2**, a source electrode **150** disposed in the first region R**1** on the upper surface of the dielectric layer **125** having corners, and a gate bus line **160** disposed in the second region R**2**. Accordingly, the power semiconductor device **100** may prevent cracks or inflow of foreign substances due to the step difference between the first region R**1** (cell region) and the second region R**2** (edge region), thereby providing power semiconductor devices having improved reliability.

[0066] According to example embodiments, at least one of the source contact plug **141** and the first peripheral contact plug **143** may include the same material as the source electrode **150**. At least one of the source contact plug **141** and the first peripheral contact plug **143** may be integrated with the source electrode **150**, and the interface between the two may not be identified. Similarly, the gate bus line **160** and the gate contact plug **145** may include the same material, and may be formed integrally, and the interface therebetween may not be discernible. Additionally, the conductive terminal **170** and the second peripheral contact plug **147** may include the same material, and may be formed as one body, and the interface between the two may not be identifiable.

[0067] FIG. **3**A is a cross-sectional view illustrating another embodiment of the power semiconductor device of FIG. **1** along line I-I'. FIG. **3**B is an enlarged view of region B of the power semiconductor device of FIG. **3**A.

[0068] Referring to FIG. **3**A, the components of a power semiconductor device **100***a*, except for the dielectric layer **125'**, the interlayer insulating layer **127**, the source contact plug **141'**, the gate contact plug **145'**, and the first and second peripheral contact plugs **143'** and **147'** may be the same as or similar to the configurations of the power semiconductor device **100** of FIG. **2**A.

[0069] The power semiconductor device **100***a* may include gate electrodes **130**, a conductive pattern **135**, a dielectric layer **125'** covering the gate electrodes **130** and the conductive pattern **135**, and an interlayer insulating layer **127** disposed on the dielectric layer **125'**. The power semiconductor device **100***a* may include a source contact plug **141'**, a gate contact plug **145'**, and first and second peripheral contact plugs **143'** and **147'** penetrating through the dielectric layer **125'** and the interlayer insulating layer **127**.

[0070] The power semiconductor device **100***a* may include a conductive pattern **135** having an upper surface disposed on the same level as the upper surface of the gate electrodes **130**. In an example, the upper surfaces of the first pattern **135***a* and the second pattern **135***b* of the conductive pattern **135** may be disposed on substantially the same level as the upper surfaces of the gate electrodes **130**.

[0071] The dielectric layer **125'** may cover the gate electrodes **130** and the conductive pattern **135**. An interlayer insulating layer **127** may be disposed on the dielectric layer **125'**.

[0072] A source electrode **150**, a gate bus line **160**, and a conductive terminal **170** may be disposed on the interlayer insulating layer **127**.

[0073] The interlayer insulating layer **127** may include an insulating material, for example, at least one of silicon oxide, silicon nitride, and silicon oxynitride.

[0074] The source electrode **150** may be connected to the source regions **107** and the well contact region **108** through a source contact plug **141'** penetrating through the dielectric layer **125'** and the interlayer insulating layer **127**. In an example, the source electrode **150** may be connected to the first peripheral contact region **109*a*** through the first peripheral contact plug **143'** penetrating through the dielectric layer **125'** and the interlayer insulating layer **127**.

[0075] The gate bus line **160** may be connected to the conductive pattern **135** through a gate contact plug **145'** penetrating through the dielectric layer **125'** and the interlayer insulating layer **127**.

[0076] The conductive terminal **170** may be connected to the second peripheral contact region **109*b*** through the second peripheral contact plug **147'** penetrating through the dielectric layer **125'** and the interlayer insulating layer **127**.

[0077] The source contact plug **141'** may include a first source contact plug portion **141*a*** penetrating through the dielectric layer **125'** and a second source contact plug portion **141*b*** penetrating through the interlayer insulating layer **127**. In an example, a side surface of the first source contact plug portion **141*a*** may be surrounded by a dielectric layer **125'**, and the first source contact plug portion **141*a*** may have a first width **W1** in the horizontal direction. A side surface of the second source contact plug portion **141*b*** may be surrounded by an interlayer insulating layer **127**, and the second source contact plug portion **141*b*** may have a second width **W2** that is larger than the first width **W1** in the horizontal direction.

[0078] The gate contact plug **145'** may include a first gate contact plug portion **145*a*** penetrating through the dielectric layer **125'** and a second gate contact plug portion **145*b*** penetrating through the interlayer insulating layer **127**. The side surface of the first gate contact plug portion **145*a*** may be surrounded by the dielectric layer **125'**, and the side surface of the second gate contact plug portion **145*b*** may be surrounded by the interlayer insulating layer **127**. In an example, the width of the second gate contact plug portion **145*b*** in the horizontal direction may be larger than the width of the first gate contact plug portion **145*a*** in the horizontal direction.

[0079] The first peripheral contact plug **143'** may include a 1-1 peripheral contact plug portion **143*a*** penetrating through the dielectric layer **125'** and a 1-2 peripheral contact plug portion **143*b*** penetrating through the interlayer insulating layer **127**.

[0080] The second peripheral contact plug **147'** may include a 2-1 peripheral contact plug portion **147*a*** penetrating through the dielectric layer **125'** and a 2-2 peripheral contact plug portion **147*b*** penetrating through the interlayer insulating layer **127**.

[0081] In an example, the side of the 1-1 peripheral contact plug portion **143*a*** and the 2-1 peripheral contact plug portion **147*a*** may be surrounded by the dielectric layer **125'**. The side surface of the 1-2 peripheral contact plug portion **143*b*** and the side surface of the 2-2 peripheral contact plug portion **147*b*** may be surrounded by the interlayer insulating layer **127**.

[0082] The dielectric layer **125'** may include a side **125'S1** surrounding the side of the first source contact plug portion **141*a*,** an upper surface **125'S2** in contact with the lower surface of the second source contact plug portion **141*b*** and the lower surface of the interlayer insulating layer **127**, and a curved area **125**RA connecting the side **125'S1** and upper surface **125'S2**. The curved area **125**RA may have a predetermined curvature.

[0083] The power semiconductor device **100*a*** according to example embodiments includes a conductive pattern **135** extending from the first region R**1** and disposed in the second region R**2** and having an upper surface disposed on the same level as the upper surface of the gate electrodes **130** disposed in the first region R**1**, and thus the step difference between the first region R**1** and the second region R**2** may be significantly reduced, thereby providing a power semiconductor device having improved reliability.

[0084] FIGS. **4**A to **4**E are cross-sectional views illustrating another embodiment of the power semiconductor device of FIG. **1** along line I-I'.

[0085] Referring to FIG. **4**A, the remaining configurations of a power semiconductor device **100*b*,** except for the dielectric layer **125**, may be the same or similar to the configurations of the power semiconductor device **100*a*** of FIG. **3**A.

[0086] The power semiconductor device **100*b*** may include gate electrodes **130**, a conductive pattern **135**, a dielectric layer **125** covering the gate electrodes **130** and the conductive pattern **135**, and an interlayer insulating layer **127** disposed on the dielectric layer **125**. In an example, the power semiconductor device **100*b*** may include a source contact plug **141'**, a gate contact plug **145'**, and first and second peripheral contact plugs **143'** and **147'** penetrating through the dielectric layer **125** and the interlayer insulating layer **127**.

[0087] The power semiconductor device **100*b*** may include a conductive pattern **135** having an upper surface disposed on the same level as the upper surface of the gate electrodes **130**.

[0088] The dielectric layer **125** may cover the gate electrodes **130** and the conductive pattern **135**. An interlayer insulating layer **127** may be disposed on the dielectric layer **125**.

[0089] A source electrode **150**, a gate bus line **160**, and a conductive terminal **170** may be disposed on the interlayer insulating layer **127**.

[0090] The source electrode **150** may be connected to the source regions **107** and the well contact region **108** through a source contact plug **141'** penetrating through the dielectric layer **125** and the interlayer insulating layer **127**. In an example, the source electrode **150** may be connected to the first peripheral contact region **109*a*** through the first peripheral contact plug **143'** penetrating through the dielectric layer **125** and the interlayer insulating layer **127**.

[0091] The gate bus line **160** may be connected to the conductive pattern **135** through a gate contact plug **145'** penetrating through the dielectric layer **125** and the interlayer insulating layer **127**.

[0092] The conductive terminal **170** may be connected to the second peripheral contact region **109*b*** through the second peripheral contact plug **147'** penetrating through the dielectric layer **125** and the interlayer insulating layer **127**.

[0093] The conductive terminal **170** may be connected to the second peripheral contact region **109*b*** through the second peripheral contact plug **147'** penetrating through the dielectric layer **125** and the interlayer insulating layer **127**.

[0094] The source contact plug **141'** may include a first source contact plug portion **141***a* penetrating through the dielectric layer **125** and a second source contact plug portion **141***b* penetrating through the interlayer insulating layer **127**.

[0095] The gate contact plug **145'** may include a first gate contact plug portion **145***a* penetrating through the dielectric layer **125** and a second gate contact plug portion **145***b* penetrating through the interlayer insulating layer **127**.

[0096] The first peripheral contact plug **143'** may include a 1-1 peripheral contact plug portion **143***a* penetrating through the dielectric layer **125** and a 1-2 peripheral contact plug portion **143***b* penetrating through the interlayer insulating layer **127**.

[0097] The second peripheral contact plug **147'** may include a 2-1 peripheral contact plug portion **147***a* penetrating through the dielectric layer **125** and a 2-2 peripheral contact plug portion **147***b* penetrating through the interlayer insulating layer **127**.

[0098] The dielectric layer **125** may include a side surrounding the side of the first source contact plug portion **141***a* (for example, side **125S1** in FIG. **2**B), and an upper surface extending from the side and contacting the lower surface of the second source contact plug portion **141***b* and the lower surface of the interlayer insulating layer **127** (for example, upper surface **125S2** in FIG. **2**B). The dielectric layer **125** may include a corner formed in an area where the side and upper surfaces come into contact.

[0099] The power semiconductor device **100***b* according to example embodiments may include a conductive pattern **135** extending from the first region R**1** and disposed in the second region R**2** and having an upper surface disposed on the same level as the upper surfaces of the gate electrodes **130** disposed in the first region R**1**, and a dielectric layer **125** having a corner. Accordingly, a power semiconductor device having improved reliability may be provided by significantly reducing the step difference between the first region R**1** and the second region R**2**.

[0100] Referring to FIG. **4**B, the remaining configurations of a power semiconductor device **100***c*, except for the dielectric layer **125'**, may be the same or similar to the configurations of the power semiconductor device **100** of FIG. **2**A.

[0101] The power semiconductor device **100***c* may include gate electrodes **130**, conductive patterns **135**, a dielectric layer **125'** covering the gate electrodes **130** and the conductive pattern **135**, and a source electrode **150**, a gate bus line **160** and a conductive terminal **170** disposed on the dielectric layer **125'**.

[0102] The source electrode **150**, gate bus line **160**, and conductive terminal **170** may be in contact with the upper surface of the dielectric layer **125'**.

[0103] The source electrode **150** may be connected to the source regions **107** and the well contact region **108** through the source contact plug **141** penetrating through the dielectric layer **125'**. In an example, the source electrode **150** may be connected to the first peripheral contact region **109***a* through the first peripheral contact plug **143** penetrating through the dielectric layer **125'**.

[0104] The gate bus line **160** may be connected to the conductive pattern **135** through the gate contact plug **145** penetrating through the dielectric layer **125'**.

[0105] The conductive terminal **170** may be connected to the second peripheral contact region **109***b* through the second peripheral contact plug **147** penetrating through the dielectric layer **125'**.

[0106] The dielectric layer **125'** may include a side surrounding the side of the source contact plug **141** (for example, side **125'S1** in FIG. **3**B), an upper surface in contact with the lower surface of the source electrode **150** (for example, upper surface **125'S2** in FIG. **3**B), and a curved area extending from the side and connected to the upper surface (for example, curved area **125**RA in FIG. **3**B). The curved area may have a predetermined curvature.

[0107] Each side of the source contact plug **141**, the gate contact plug **145**, and the first and second peripheral contact plugs **143** and **147** may be surrounded by a dielectric layer **125'**.

[0108] The upper surface of the source contact plug **141** may be disposed on the same level as the upper surface of the dielectric layer **125'**. In an example, the upper surface of the gate contact plug **145** and the upper surfaces of the first and second peripheral contact plugs **143** and **147** may be disposed on the same level as the upper surface of the dielectric layer **125'**.

[0109] The power semiconductor device **100***c* according to example embodiments includes a conductive pattern **135** extending from the first region R**1** and disposed in the second region R**2**, and having an upper surface disposed on the same level as the upper surface of the gate electrodes **130** disposed in the first region R**1**, a source contact plug **141** having an upper surface disposed on the same level as the upper surface of the dielectric layer **125'**, a gate contact plug **145**, and first and second peripheral contact plugs **143** and **147**. Therefore, the step difference between the first region R**1** and the second region R**2** may be significantly reduced, thereby providing power semiconductor devices having improved reliability.

[0110] Referring to FIG. **4**C, the remaining components of a power semiconductor device **100***d*, excluding the conductive pattern **135'**, the interlayer insulating layer **127**, the source contact plug **141'**, the gate contact plug **145'**, and the first and second peripheral contact plugs **143'** and **147'** may be the same or similar to the configurations of the power semiconductor device **100** of FIG. **2**A.

[0111] The power semiconductor device **100***d* may include gate electrodes **130**, a conductive pattern **135'**, a dielectric layer **125** covering the gate electrodes **130** and the conductive pattern **135'**, and an interlayer insulating layer **127** disposed on the dielectric layer **125**. In an example, the power semiconductor device **100***d* may include a source contact plug **141'**, a gate contact plug **145'**, and first and second peripheral contact plugs **143'** and **147'** penetrating through the dielectric layer **125** and the interlayer insulating layer **127**.

[0112] The conductive pattern **135'** may be disposed on a portion of the insulating liner **121** and the insulating pattern **123** and spaced apart from the gate electrodes **130**. In an example, the conductive pattern **135'** may include a first pattern **135***a* in contact with the upper surface of the insulating liner **121**, and a second pattern **135***b'* extending from the first pattern **135***a* and in contact with a portion of the insulating pattern **123**.

[0113] The height of the first pattern **135***a* in the vertical direction (Z-direction) from the upper surface of the insulating liner **121** may be substantially equal to the height of the second pattern **135***b*' in the vertical direction (Z-direction) from the upper surface of the insulating pattern **123**.

[0114] The upper surface of the first pattern **135***a* may be disposed at a lower level than the upper surface of the second pattern **135***b*'. In an example, the upper surface of the second pattern **135***b*' may be disposed at a level higher than the upper surface of the first pattern **135***a* by the height of the insulating pattern **135***b*' in the vertical direction (Z-direction).

[0115] The upper surface of the first pattern **135***a* may be disposed on the same level as the upper surface of the gate electrode **130**. In an example, the upper surface of the second pattern **135***b*' may be disposed at a higher level than the upper surface of the gate electrode **130**.

[0116] The dielectric layer **125** may cover the gate electrode **130** and the conductive pattern **135**'.

[0117] The power semiconductor device **100***d* according to example embodiments includes a dielectric layer **125** having corners, thereby significantly reducing the step difference between the first region R**1** and the second region R**2** and thus providing power semiconductor devices having improved reliability.

[0118] Referring to FIG. **4**D, the remaining configurations of a power semiconductor device **100***e*, except for the conductive pattern **135**', may be the same or similar to those of the power semiconductor device **100** of FIG. **2**A.

[0119] The power semiconductor device **100***e* according to example embodiments includes a dielectric layer **125** having a corner, a source contact plug **141**, a gate contact plug **145** and first and second peripheral contact plugs **143** and **147** having an upper surface disposed on the same level as the upper surface of the dielectric layer **125**, thereby significantly reducing the step difference between the first region R**1** and the second region R**2** and thus providing a power semiconductor device having improved reliability.

[0120] Referring to FIG. **4**E, the remaining configurations of a power semiconductor device **100***f*, except for the conductive pattern **135**' and the dielectric layer **125**', may be the same as or similar to those of the power semiconductor device **100** of FIG. **2**A.

[0121] The power semiconductor device **100***f* according to example embodiments includes a source contact plug **141**, a gate contact plug **145**, and first and second peripheral contact plugs **143** and **147** having an upper surface disposed on the same level as the upper surface of the dielectric layer **125**', thereby significantly reducing the step difference between the first region R**1** and the second region R**2** and thus providing power semiconductor devices having improved reliability.

[0122] FIGS. **5**A to **5**F are diagrams illustrating an example embodiment of the method of manufacturing the power semiconductor device of FIG. **2**A.

[0123] Referring to FIG. **5**A, a method of manufacturing method a power semiconductor device may include forming a drift layer **103** on the substrate **101**, and forming first and second well regions **105***a* and **105***b*, source regions **107** and well contact regions **108** in the first well region **105***a* and first and second peripheral contact regions **109***a* and **109***b* within second well region **105***b*.

[0124] The manufacturing method may include forming an insulating liner **121** on the first and second well regions **105***a* and **105***b*, and forming an insulating pattern **123** on the insulating liner **121** formed on the second well region **105***b*.

[0125] The manufacturing method may include forming gate electrodes **130** spaced apart from each other in the first direction (X-direction) with the source regions **107** interposed therebetween, on the insulating liner **121** formed on the first well region **105***a*. The manufacturing method may include forming a conductive pattern **135**' spaced apart from the gate electrodes **130** and covering at least a portion of the insulating liner **121** and the insulating pattern **123** formed on the second well region **105***b*.

[0126] The conductive pattern **135**' may include a first pattern **135***a* in contact with the upper surface of the insulating liner **121** formed on the second well region **105***b*, and a second pattern **135***b*' in contact with the upper surface of the insulating pattern **123**. The heights of the first pattern **135***a* and the second pattern **135***b*' in the vertical direction may be substantially the same as each other.

[0127] In an example, the upper surface of the first pattern **135***a* of the conductive pattern **135**' may be disposed on substantially the same level as the upper surface of the gate electrode **130**.

[0128] The conductive pattern **135**' may be formed in the same process as the gate electrodes **130**. In an example, the gate electrodes **130** and conductive pattern **135**' may be deposited using deposition techniques such as chemical vapor deposition (CVD), plasma enhanced chemical vapor deposition (PE-CVD), or spin on dielectric (SOD).

[0129] Referring to FIG. **5**B, the manufacturing method may include forming a conductive pattern **135** by performing a chemical mechanical polishing (CMP) process to flatten the conductive pattern **135**. The conductive pattern **135** may be formed by etching the second pattern **135***b* of the conductive pattern **135** in the vertical direction (Z-direction) to have the same level as the upper surface of the first pattern **135***a*. In an example, upper surfaces of the first pattern **135***a* and the second pattern **135***b* of the conductive pattern **135** may be coplanar.

[0130] Referring to FIG. **5**C, the manufacturing method may include removing a portion of the insulating liner **121** to expose a portion of the source regions **107**, the well contact region **108**, and the first and second peripheral contact regions **109***a*, **109***b*.

[0131] The manufacturing method may include forming parts of source regions **107** and well contact region **108**, and metal-semiconductor compound layers **152***a*, **152***b*, and **152***c* on the first and second peripheral contact regions **109***a* and **109***b*, respectively.

[0132] The method may include forming a dielectric layer **125**' covering the gate electrodes **130** and the conductive pattern **135**. The metal-semiconductor compound layers **152***a*, **152***b*, and **152***c* may be exposed through the openings OPN formed in the dielectric layer **125**'.

[0133] The dielectric layer **125**' may include a side of the openings OPN (for example, side **125**'S**1** in FIG. **3**B) and an upper surface extending from the side (for example, upper surface **125**'S**2** in FIG. **3**B), and may include a curved area connecting the side surface to the upper surface (for example, the curved area **125**RA in FIG. **3**B).

[0134] Referring to FIG. **5**D, the step difference of forming a flattened dielectric layer **125** by performing a chemical mechanical polishing (CMP) process to remove the curved area of the dielectric layer **125**' may be included. The dielectric layer **125**' may be formed as a dielectric layer **125** having corners through the CMP process.

[0135] Referring to FIG. **5**E, forming plug structures **140***a* and **140***b* that fill the openings OPN may be included.

[0136] The plug structures **140***a* and **140***b* may include a first plug structure **140***a* filling the first and second openings OPN**1** and OPN**2**, and second plug structures **140***b* filling the third and fourth openings OPN**3** and OPN**4**.

[0137] The first plug structure **140***a* may include a source contact plug **141** filling the first opening OPN**1** exposing the first metal-semiconductor compound layer **152***a* disposed on the source regions **107** and the well contact region **108**, a first peripheral contact plug **143** filling the second opening OPN**2** exposing the second metal-semiconductor compound layer **152***b* disposed on the first peripheral contact region **109***a*, and a first conductive structure **142** formed on the dielectric layer **125** and integrated with the source contact plug **141** and the first peripheral contact plug **143**.

[0138] The second plug structure **140***b* may include a gate contact plug **145** filling the third opening OPN**3** exposing a portion of the conductive pattern **135**, a second peripheral contact plug **147** filling the fourth opening OPN**4** exposing the third metal-semiconductor compound layer **152***c* disposed on the second peripheral contact region **109***b*, and a second conductive structure **146** formed on the dielectric layer **125** and integrated with the gate contact plug **145** and the second peripheral contact plug **147**.

[0139] The first and second plug structures **140***a* and **140***b* are illustrated to be formed spaced apart in the first direction (X-direction), but are not limited thereto. The first plug structure **140***a* and the second plug structure **140***b* may be formed integrally. The first and second plug structures **140***a* and **140***b* may be formed through a plating process, a PVD process, or a CVD process.

[0140] Referring to FIG. **5**F, a CMP process may be performed to remove the first conductive structure **142** and the second conductive structure **146** formed on the dielectric layer **125**.

[0141] By the CMP process, upper surfaces of the source contact plug **141**, the gate contact plug **145**, and the first and second peripheral contact plugs **143** and **147** may be formed to have the same level. In an example, the upper surfaces of the gate contact plug **145** and the first and second peripheral contact plugs **143** and **147** may be formed to have the same level as the upper surface of the dielectric layer **125**.

[0142] A method of manufacturing a power semiconductor device according to example embodiments may include a planarization process operation for the conductive pattern **135** formed over the first region R**1** and the second region R**2**, a planarization process operation for the dielectric layer **125** covering the gate electrodes **130** and the conductive pattern **135**, and a planarization process operation for the source contact plug **141** to electrically connect the source electrode **150** formed on the dielectric layer **125** to the source regions **107**, and thus, the step difference between the first and second regions R**1** and R**2** may be removed, thereby providing power semiconductor devices having improved reliability.

[0143] As set forth above, power semiconductor devices according to example embodiments may include a conductive pattern, a dielectric layer, and a contact plug disposed in a connection region extending from the cell region, and at least one of the conductive pattern, the dielectric layer, and the contact plug may be planarized by a polishing process. Accordingly, the power semiconductor device may have improved reliability by improving the step difference between a cell region and the connection region.

[0144] While example embodiments have been illustrated and described above, it will be apparent to those skilled in the art that modifications and variations could be made without departing from the scope of the appended claims.

What is claimed is:

1. A power semiconductor device comprising:

a substrate including a cell region and an edge region extending from the cell region;

a drift layer of a first conductivity type on the substrate;

a first well region of a second conductivity type extending from an upper surface of the drift layer in the cell region and further disposed within the drift layer;

a source region of the first conductivity type extending from an upper surface of the first well region and further disposed within the first well region;

an insulating liner on the drift layer;

gate electrodes spaced apart from each other on the insulating liner in the cell region, with the source region interposed between the gate electrodes;

an insulating pattern in the edge region;

a conductive pattern spaced apart from the gate electrodes and covering a portion of the insulating liner and further covering a portion of the insulating pattern;

a dielectric layer covering the gate electrodes and further covering the conductive pattern;

a source electrode on the dielectric layer; and

a source contact plug penetrating through the dielectric layer and connecting the source electrode and the source region,

wherein the conductive pattern includes a first pattern in contact with the insulating liner and a second pattern in contact with the insulating pattern, and an upper surface of the first pattern is disposed on a same level as an upper surface of the second pattern.

2. The power semiconductor device of claim **1**, wherein an upper surface of the dielectric layer is disposed on a same level as an upper surface of the source contact plug.

3. The power semiconductor device of claim **1**, wherein the source electrode contacts a portion of an upper surface of the dielectric layer and further contacts an upper surface of the source contact plug.

4. The power semiconductor device of claim **1**, further comprising an interlayer insulating layer on the dielectric layer,

wherein the source contact plug includes a first source contact plug portion surrounded by the dielectric layer and a second source contact plug portion extending from the first source contact plug portion and surrounded by the interlayer insulating layer.

5. The power semiconductor device of claim **4**, wherein a width of the first source contact plug portion in a first direction is smaller than a width of the second source contact plug portion in the first direction.

6. The power semiconductor device of claim **1**, wherein the dielectric layer includes a side in contact with the source contact plug, an upper surface extending from the side, and a corner where the side and the upper surface are in contact.

7. The power semiconductor device of claim **1**, wherein the dielectric layer includes a side in contact with the source contact plug, an upper surface extending from the side, and a curved area between the side and the upper surface.

8. The power semiconductor device of claim **1**, further comprising:

a gate bus line spaced apart from the source electrode and disposed in the edge region; and

a gate bus line contact plug penetrating through the dielectric layer and connecting the gate bus line and the conductive pattern.

9. The power semiconductor device of claim **8**, wherein a lower surface of the source electrode is disposed on a same level as a lower surface of the gate bus line.

10. The power semiconductor device of claim **8**, wherein the gate bus line contact plug overlaps the second pattern in a vertical direction.

11. The power semiconductor device of claim **1**, further comprising a well contact region of the second conductivity type extending from the upper surface of the first well region, disposed within the first well region, and crossing the source region.

12. The power semiconductor device of claim **1**, further comprising a second well region of the second conductivity type spaced apart from the first well region, extending from the upper surface of the drift layer in the edge region, and disposed within the drift layer,

wherein the conductive pattern overlaps the second well region.

13. The power semiconductor device of claim **12**, further comprising guard rings of the second conductivity type spaced apart from the conductive pattern in a first direction, extending from an upper surface of the second well region, and disposed in the second well region.

14. A power semiconductor device comprising:

a substrate including a first region and a second region surrounding the first region;

a drift layer of a first conductivity type on the substrate;

a first well region of a second conductivity type extending from an upper surface of the drift layer in the first region and further disposed within the drift layer;

a source region of the first conductivity type extending from an upper surface of the first well region and further disposed within the first well region;

an insulating liner on the drift layer;

an insulating pattern in the second region;

gate electrodes on the insulating liner in the first region and spaced apart from each other in a first direction with the source region interposed between the gate electrodes;

a conductive pattern spaced apart from the gate electrodes and covering a portion of the insulating liner and further covering a portion of the insulating pattern;

a dielectric layer covering the gate electrodes and further covering the conductive pattern; and

a source electrode on the dielectric layer in the first region,

wherein the conductive pattern includes a first pattern disposed on the insulating liner and a second pattern extending from the first pattern and disposed on the insulating pattern, and

the first pattern has a first height in a vertical direction intersecting the first direction from an upper surface of the insulating liner, and the second pattern has a second height, shorter than the first height, in the vertical direction, from an upper surface of the insulating pattern.

15. The power semiconductor device of claim **14**, wherein upper surfaces of the gate electrodes are disposed on a same level as an upper surface of the conductive pattern.

16. The power semiconductor device of claim **14**, wherein upper surfaces of the gate electrodes are disposed on a level higher than the upper surface of the insulating pattern.

17. The power semiconductor device of claim **14**, wherein an upper surface of the first pattern is disposed on a same level as an upper surface of the second pattern.

18. The power semiconductor device of claim **14**, wherein the conductive pattern includes a same material as the gate electrodes.

19. A power semiconductor device comprising:

a substrate of a first conductivity type;

a drift layer of the first conductivity type on the substrate;

a first well region of a second conductivity type on the drift layer;

a second well region of the second conductivity type disposed on the drift layer and spaced apart from the first well region in a first direction;

source regions of the first conductivity type on the first well region;

an insulating liner on the drift layer;

gate electrodes spaced apart from each other, with a source region of the source regions interposed between the gate electrodes, on the first well region;

an insulating pattern disposed on a portion of the insulating liner, on the second well region;

a conductive pattern spaced apart from the gate electrodes and covering a portion of the insulating liner and further covering a portion of the insulating pattern on the second well region;

a dielectric layer covering the gate electrodes and the conductive pattern;

a source electrode on the dielectric layer; and

a source contact plug penetrating through the dielectric layer and connecting the source electrode and the source region,

wherein the conductive pattern includes a first pattern in contact with the insulating liner and a second pattern in contact with the insulating pattern, and an upper surface of the first pattern and an upper surface of the second pattern are disposed on a same level, and

an upper surface of the dielectric layer is disposed on a same level as an upper surface of the source contact plug.

20. The power semiconductor device of claim **19**, wherein the dielectric layer includes a side in contact with the source contact plug, an upper surface extending from the side, and a corner where the side and the upper surface are in contact.

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